

Consider what the cost price is made up of, such as material, machine, operator, overhead, storage, assembly costs, etc. Don't forget to include the design costs as well.

Poster: we need to recognise our mistakes and put them in our poster and understand how we can improve

[illegible]

# units	Cost price/unit	Total price
800000	0,0785	62800
		0
		0
200000		0
		0
22223		0
		0
200000		0
		0
		0
		0
		0
		0
		0
400000 (grams)	0.0015 euros	6000
		0
		0
		0
		0
		0
		0
		0

We need 30x30 plates for our car vacuum form

Introduction

Why is a cost analysis important

A cost analysis for mass production is crucial, which will be made in the project in order to help the company to study the costs and expenses of the parts for the desired number products consisting of different parts—like sub-assemblies such as wheels, shafts and the bodies—.

It serves as a guidance for the project to identify in a short term, the areas and the steps whereas expenses can be minimized without affecting the quality of product, on a long term, the sales trend of *Speed Mass*. A closer look at the market and mass-produced toy cars helps in setting competitive prices by finding a balance between appropriate pricing and market competitiveness. By looking into the ordered parts prices and the needed processes with what materials will be used, cost analysis will navigate if there are any shortages, figure out the unnecessary components that would increase the efficiency of the mass production

Scope of the analysis

Cost analysis will be done by importing a table of ordered parts which contains each sub assembly, and the process needed and the overall cost. In addition to the discussion of the materials, machining costs, operator, overheads, storage cost, and assembly costs.

Cost calculation methodology

Material cost

The wheels were ordered from Cube production. All wheels are made of polypropylene which is a thermoplastic polymer, it is supplied as an injection molded product as set of 4 with runner system and sprue attached.

According to the market, polypropylene is 0.0015 euros per gram. An estimation for the overall weight of the car would be 20 grams for a car that is hollow from the inside. Hence to find how much polypropylene costs will be obtained by first computing the total polypropylene needed in grams.

$$\text{Total polypropylene needed} = \text{Weight per frame} \times \text{Number of frames}$$

$$= 20g \times 200,000 = 4,000,000\text{grams}$$

Finding the total cost of plastic will be calculated by the total polypropylene multiplied by the 0.0015, hence the equation will be as following:

$$\text{total cost of the polypropylene} = \text{price per gram} \cdot \text{total propylene needed}$$

$$= 4,000,000 \times 0.0015 = 6,000\text{€}$$

The shaft will be made from steel.

The material of the body will PS (white) with a thickness of 2 mm.

The material of the frame will be PP (SABIC 576P).

The mold will be CNC milled out of aluminum, the mold block is 139,9 mm in width, 159,9 mm in length and 24,6 mm thick. The material of the bottom plate is steel.

Machine(s)

The machines that will be used in this process are injection molding machines, vacuum forming machines, 3D printers and milling machines. Each one has set-up cost, which is usually calculated by hour, and labour cost per hour (which could potentially be optimised depending on the machinery cost).

The overall cost for the machine (both milling machine and the injection moulding machine) is the sum of the depreciation cost, labor cost and the CAM cost.

What is the depreciation cost?

Depreciation cost of the machine stands for the reduction cost after the machine life is complete, hence, it is calculated by:

$$\text{Depreciation per hour} = \frac{\text{purchase of the machine (€)}}{\text{estimated life (year)} \cdot \text{working hours per year}}$$

$$\text{Depreciated Cost} = \text{Depreciation per hour} \times \text{Number of Working Hours Used}$$

The purchase price of the machines differs from the milling machine, injection moulding and vacuum forming. Estimated tool life is the total useful life of each machine that will be used for producing the car frame in hours. The working hours estimates how many hours the machine has already been using

Labour cost analysis is crucial because it helps to identify which areas need to be optimized. Since it is relevant to the production line, it opens the door for many observations such as analyzing how much time is spent on bottlenecks in the production lines, and opportunities for improvement.

Labour cost per hour is identified by the employee cost per hour multiplied by the number of employees

$$\text{Labour cost per hour} = \text{employee cost per hour} \cdot \text{number of employees}$$

Vacuum forming

Injection moulding

Injection molding machine cost is calculated by the material cost, primary machine cost, secondary machine cost and overhead cost all added to each other. Material cost refers to the total expense incurred to acquire the raw materials including the plastic and the mold needed to manufacture the toy car mold and to carry out the process of the injection molding. For carrying on the calculations, material cost formula is computed

$$\text{Material cost} = \frac{\text{plastic weight per product (g)}}{\text{material cost per gram } \left(\frac{\text{€}}{\text{g}}\right)}$$

The primary machine cost means the initial investment cost of the main injection molding machine itself. It also counts for the main components used in injection moulding such as the moulding unit, the mould, the clamping unit and the sprue by which the plastic is injected to the mould. The primary cost is the filling cost added to the cooling cost.

$$\text{primary injection moulding machine cost} = \text{filling cost} + \text{cooling cost}$$

Filling cost refers to the portion of the total production cost that is associated with filling the mold cavity with molten plastic during the injection phase of the molding cycle. Hence, the total filling cost is calculated by

$$\text{filling time (hours)} \cdot \text{injection moulding machine cost per hour}$$

Same applies with the cooling cost, to be able to compute the overall cooling cost, it would be calculated by the cooling cost (obtained through moldex) multiplied by the machine's cost per hour.

Injection moulding machine Cost (per Product): €0.2590

The injection molding machine cost is broken down into sub-analysis according to the spreadsheet

Material Cost: €0.0296

Based on plastic weight per product 19.7 g and the plastic price per gram is 0.0015 €

Calculation:

$$19.7g \cdot 0.0015 \frac{\text{€}}{g} = 0.02955 \approx 0.02996$$

Primary Machine Cost: €0.1581

In the primary machine cost, 2 sub-costs were introduced in injection moulding which are the filling costs and cooling costs.

1. Filling Cost: €0.0079

Filling time calculated in hours to be **0.000321 hours** and the machine cost/hour is **€24.5731**

2. Cooling Cost: €0.1502

a. Cooling time: 0.006111 hours

The secondary machine cost refers to the cost of supporting equipment and auxiliary machines required to operate the injection molding system efficiently. These are not directly part of the injection unit but are necessary for the full production process. Due to the lack of knowledge on to what extent is the injection molding system in this case of the project is fulfilled, it would be less feasible to say which secondary costs are included, but usually secondary costs involve the mold temperature components and conveyors. Secondary machine cost is calculated by secondary time taken by the machine (in hours) (to cool down, to clean, clamp) multiplied by the injection moulding machine cost per hour.

Overheads cost is calculated by the overhead cost per hour multiplied by the addition of the primary and secondary costs

$$\text{Overheads cost} = \text{overhead cost per hour} \cdot (\text{primary} + \text{secondary cost})$$

Evaluation

Assembly costs and packaging cost

For the assembly costs, it is the sum of the final cost per product and the labour cost.

Production

Objective

The company **SpeedMass** aims to optimize its production layout to reduce both cost and time. Before diving into details with **FlexSim**, key performance indicators (KPIs) were defined to guide the project lifecycle. Several KPIs focusing on cost and production efficiency include:

- **Lagging KPI:** Product return rate (lower return rates indicate higher product quality and customer satisfaction)
- **Lagging KPI:** Monthly units sold (reflects sales performance)
- **Leading KPI:** Time required to produce one unit (indicates production line efficiency). Helps to improve the performance of the next car produced.

Production Line Optimization

To optimize production costs, it is also necessary to improve the efficiency of the production line. Using FlexSim software, several analyses were conducted—ranging from determining the optimal number of workstations to identifying the most efficient quantity of materials to keep in inventory. Additionally, the analysis considered that increasing the speed of the production line may require hiring more workers, which represents an additional investment.

To create the simulations, the takt time was calculated first:

$$\text{Takttime: } \frac{\text{available production time}}{\text{customer demand}} = \frac{8 \cdot 209 \cdot 3600}{200000} = 30.096 \text{ seconds/unit}$$

- It is assumed that 200,000 units will be sold to the company.
- Based on average working hours in the Netherlands (where SC-04, our product, will be manufactured), it is assumed that the production line will operate 8 hours a day, 5 days a week. And, it is assumed that the worker has 20 paid vacation days and 11 national holidays (which in the worst case scenario are all during the week) so in total, 209 workable days.
- At this stage, the possibility of having two shifts is not considered.

This calculation implies that no workstation should take longer than 30.096 seconds per unit, as this would create a bottleneck in the production line. Since the longest process is the injection molding (30 seconds) and is slightly under the takt time, there are no bottlenecks.

Following a thorough evaluation, the production process was broken down into the following workstations (WS):

- **WS1:** Fixing two wheels to the axle (press-fitted twice). Estimated process time: 15 seconds
- **WS2:** Attaching the axles (with wheels) to the frame and put on the screws. Estimated process time: 20 seconds
- **WS3:** Assembling the frame (with axles) to the bottom plate using screws. Estimated process time: 20 seconds
- **WS4:** Snapping the body onto the frame and performing a quality check. Estimated process time: 15 seconds

With this information, we can now know the cycle time, which is going to be 30 seconds due to the injection moulding that is the longest process. This means that every 30 seconds one product will be manufactured. Even though, the total processing time will be $30+15+20+20+15=100$ seconds.

To meet production goals efficiently, the cycle time (how long it takes to produce one unit) should align with the takt time (the maximum time allowed per unit to meet customer demand). In this case, the demand can be met because the takt time (30.096 seconds) is slightly higher than the cycle time (30 seconds). This was considered the worst-case scenario during planning.

If we wanted to increase the gap between cycle time and takt time, one option could be to run the injection moulding machine during the night. This would ensure that parts are pre-produced before regular working hours. However, since this process is not currently a bottleneck (based on our assumptions), there is no need to do so.

Moreover, reducing the cycle time even further could lead to overproduction, which goes against lean manufacturing principles.

On the other hand, there is a significant chance that some units may be defective. To account for this, producing slightly more than the required quantity—about 0.5% extra—could be a smart strategy. For example, if the original demand is 200,000 toy cars, this adjustment would raise the production target to 201,000 units to ensure delivery even in case of defects.

$$\text{New takt time} = \frac{\text{available production time}}{\text{customer demand}} = \frac{8 \cdot 209 \cdot 3600}{201000} = 29.496 \text{ seconds}$$

Since injection moulding represents a bottleneck in the process, operating the machine overnight could be a viable strategy to pre-produce components and prevent delays during daytime operations.

Conclusion on Production Line Optimization

The production line optimization focused on aligning the cycle time with the takt time to ensure customer demand is met efficiently and consistently.

A detailed analysis of individual workstations and task durations revealed opportunities to balance workloads and implement parallel operations, thereby reducing overall cycle time and mitigating bottlenecks. Furthermore, introducing a small production buffer of approximately 0.5–1% accounts for potential defects without leading to significant overproduction.

These optimization efforts significantly enhance the responsiveness, reliability, and efficiency of the production process, while supporting key performance indicators (KPIs) related to cost control, product quality, and throughput. By aligning operations with these KPIs, the production line is better equipped to deliver high performance and customer satisfaction while minimizing waste.

Storage optimization

It was decided to store raw materials within the factory premises, as this approach offers several operational and financial advantages aligned with cost optimization.

For SpeedMass, immediate accessibility to raw materials is crucial to ensure that production remains within the time constraints defined by the takt time. As such, materials will be stored on-site in the main inventory area, rather than in a separate location with additional utility and rental costs. This proximity significantly reduces lead time between material retrieval and production, enabling a smoother workflow and minimizing delays.

If off-site storage were chosen, several challenges would arise, including:

- Additional transportation costs
- Time delays due to transit and potential traffic issues

- Increased logistical complexity
- Greater risk of production disruption

By eliminating the need for third-party warehousing or external storage, SpeedMass reduces fuel consumption, minimizes handling delays, and improves production reliability.

Moreover, on-site storage enhances coordination between inventory and production departments. It enables real-time inventory tracking, more accurate stock management, reduced overstocking, and fewer interruptions due to material shortages.

FlexSim Simulation

On the other hand, what wants to be found in FlexSim simulation are factors like:

- Throughput: how many products are completed

$$970 \cdot 209 = 202730$$
- Average Cycle Time
- Queue length
- Operator Idle Time (to reduce unnecessary labour cost) $\frac{\text{idle time}}{\text{total shift time}} \cdot 100 (\%)$
- Machine utilization (check how efficiently machines are used) $\frac{\text{processing time}}{\text{available time}} \cdot 100 (\%)$

To analyze and optimize the toy car production process, a digital model of the assembly line was developed using FlexSim simulation software. The model replicates a realistic factory layout, including material sources, manufacturing machines, workstations (WS1–WS4), operators, and final output areas. Each component in the simulation corresponds to a step in the real-life assembly process.

The production line begins with three material sources —Src_Body, Src_BottomPlate, and Src_Frame— which feed into manufacturing machines: the Vacuum Former, Laser Cutter, and Injection Molding machine, respectively. These machines produce intermediate parts: the car body, bottom plate, and frame.

Parts are then queued and passed to a series of four workstations (the ones described on the section 2.2.

Operators are assigned to each station, reflecting the human effort involved. Finished cars are collected at the FinishedCars output station.

This model enables the team to test layout changes, task reassignments, and takt time alignment in a risk-free environment

- COmparision to real situtation:
 - Comparison of estimated times--> proposal of new distribution

After conducting the real-life simulation—where some team members acted as assembly line workers—several new insights were discovered. The most notable finding was that the actual task durations differed from the initial estimates.

For instance, Workstation 4 (WS4) took significantly less time than expected (approximately 4 seconds), and the associated quality check also appeared to require minimal time. Meanwhile, Workstation 3 (WS3) turned out to be the longest step, though its duration roughly matched the original estimate.

To optimize the workflow based on these observations, the team decided to redistribute tasks. Specifically, the final task was moved from WS4 to WS3, and half of WS3's original task was shifted to Workstation 2 (WS2). The new proposed distribution is:

- **WS1:** Fixing two wheels to the axle (press-fitted twice).

Estimated process time: 15 seconds

- **WS2:** Attaching the axles (with wheels) to the frame and tightening one of the bolts to assemble the frame (with axles) to the bottom plate.

Estimated process time: 20 seconds

- **WS3:** Tightening the second bolt and snap-fitting the car body onto the rest.

Estimated process time: 20 seconds

This revised setup ensures that all workers are fully utilized without idle time. Additionally, it offers cost savings, as fewer workers would be needed.

Additionally, the injection molding process was performed. However, due to the absence of ejector pins on the car frame, the process took significantly longer than expected—approximately 94 seconds. Despite this delay, it was recognized as a solvable issue that had simply not been addressed yet. Therefore, the originally estimated duration of 30 seconds will still be used in planning. It's also important to note that without the addition of ejector pins, a significant bottleneck would have arisen in the production line, further justifying the need for this design improvement.

- Mention the injection molding situation

Analysis and findings

Evaluation and recommendation

This section of the report focuses on evaluating the cost estimation strategy used in the production of the SpeedMass toy car. Several types of cost estimations were applied, including machine costs, tooling costs, and filling costs associated with the injection molding process. Additionally, vacuum forming and mould costs were considered. Beyond these, other essential expenses such as material costs, labour, packaging, and assembly were also taken into account. Together, these estimations provided a rough approximation of the final product cost.

The cost analysis was conducted as part of the recent progress in the development of SpeedMass. Understanding the company's current position relative to competitors in the market is essential. This insight will be critical in shaping future strategies to enhance the product and reduce the risk of financial loss. Moreover, by knowing the limitations of the strategy, Profit margin and markup (percentage difference between the cost of producing a good and its selling price) can be calculated

Key Assumptions

The cost estimation that was done underwent several assumptions, such as estimated lifespan for injection molding machines, vacuum forming machine and lasercutting

machine. When consulting the employees in the cube, it was mentioned that the minimum life span for the machines was 10 years.

Secondly, tooling cost for the machines was estimated, as well as the secondary time (like cleaning, clamping, etc).

Moreover, all of the overhead costs were also estimated. For vacuum forming, cooling time was estimated and for the lasercutting, the cutting time was also estimated, it was challenging to estimate this cost information due to the leakage of the softwares used in estimating these formulas.

The market research for the purchase of the machines was not accurate and the cost of one single car is not like what SpeedMass ended up being.

However, all the costs that were assumed, affected a small part of the total manufacturing cost.

A bigger picture on the cost estimation strategy used and uncertainties

Overall, there was a logical numerical consistency in the values obtained, including the costs of the machines, tooling, and cooling costs. Injection molding costs and the vacuum forming cost were in the same range of the total costs, as the same methodology for calculating the total cost was used. The end costs of the machines and the materials make sense within the context of the market. The data used in the excel sheet for the machine purchase and tooling cost were computed according to the year they were purchased. This keeps the costs of the machines up to date and accurate to the actual cost.

Vacuum forming shows a relatively lower cost compared to the other manufacturing processes. This difference is primarily attributed to the assumed cost of the other manufacturing processes. However, this assumption introduces a degree of uncertainty for the company into the overall cost estimations and may affect the accuracy of the rest of the analysis.

Breakeven analysis

Breakeven analysis is a financial tool to determine the point at which the toy car covers its costs and begins to generate profit. Bearing in mind that all the products are sold would help in estimating whether the final toy car price brings profit or loss. This is a crucial part of the evaluation process because the breakeven point would be the level of output at which total

revenue equals total costs (both fixed and variable). At this point, there is no profit or loss. It is crucial to have such input because this would put a selling goal for the SpeedMass.

The formula for the breakeven is

$$\text{Break – even point (unit)} = \frac{(\text{Fixed cost})}{\text{selling price per unit} - \text{variable cost per unit}}$$

The fixed costs in this case would be the total cost for mass production that was obtained from the sum of the machines, tooling costs, etc, which would be 631102,9426 euros.

If the business costs do not fluctuate and they are stable (however, that might be a rare condition considering the utilities for the factory, gas, electricity, etc), would make the variable cost equal to zero.

Therefore, applying the formula for the breakeven would be

$$\text{Break – even point} = \frac{631102,9426}{3.15}$$

This means that SpeedMass has to sell 200,000 toy cars to be able to get 1261102.94 euros as revenue. The revenue per unit is the total revenue obtained from the breakeven analysis divided by the total number of products that are being sold.

$$\text{Revenue per product} = \frac{1261102.94}{200000} = 6.31\text{€}$$

With these calculations it is concluded that the profit margin for SpeedMass is 50.04%. In another words, for every 1 euro of revenue, 0.50€ are kept as a profit.

$$\text{Profit} = \text{selling price} \cdot \frac{(\text{profit margin})}{100} = 3.15 \cdot \frac{(50.04)}{100} = 1.57\text{€}$$

Therefore, for each product sold, €1.57 is recorded as profit. However, it is important to note that all costs, as well as the selling price, are based on estimates. Consequently, it is not possible to determine with certainty what the actual profit will be.

While a profit margin of €1.57 per unit appears reassuring and promising at first glance, there are several significant risks that must be taken into consideration. Since the calculations rely on assumptions, any underestimation of costs, such as labor and overhead could result in a lower actual profit per unit or even lead to losses.

Furthermore, certain potential costs were not included in the cost estimation, such as shipping, marketing, and returns. These factors can substantially impact the final cost of SpeedMass products.

Relying on a €1.57 profit per unit is inherently risky, as this figure assumes that SpeedMass will sell 200,000 units, a key premise established in the earlier analysis. This sales volume reflects high demand and benefits from economies of scale. However, if demand decreases, it may ultimately incur a financial loss.

Recommendation

Following the discussion of risks and bottlenecks in the evaluation sections, it is essential to outline recommendations as the next step. The recommendations will primarily focus on implementing a failure check, incorporating a 1.5% quantity in addition to enhance reliability and mitigate the risk of losses, and accounting for costs such as marketing expenses and return-related costs. These measures can contribute to developing a more accurate cost estimate for toy cars.

Failure check

Failure check is reviewing manufacturing processes, like injection molding and vacuum forming and in addition to the assumptions that have been made to identify where failures (errors, defects, or unexpected costs) and potential bottlenecks might occur, and planning accordingly.

Therefore, a systematic failure check should be conducted in the future to assess potential points of error or risk throughout the production and sales process. One of the important steps that should be taken into consideration during the failure check is evaluating product quality to identify and minimize defect rates that could lead to returns.

This process can be done by the (IQC) or incoming quality control. This process is inspecting the raw materials that SpeedMass orders, such as the plastic sheets for the vacuum forming, axels, materials for the mold, etc. They get checked before the production starts by performing random sampling of the ordered patches and setting measurement tools to verify the dimensions and the materials.

Secondly, an additional process should be implemented accordingly, namely In-Process Quality Control (IPQC). This process involves systematic monitoring and inspection during production to enable early detection of defects.

For example, during the injection molding process, a defect occurred due to a random error in the Moldex software, which led to incorrect pressure readings. The recorded pressure required to inject the plastic into the mould was lower than the actual pressure required. After recalibration, the machine was able to inject the plastic into the mould correctly; however, some material losses were incurred.

Another instance relates to the mould design itself. The moulds should have been fabricated with additional ejection holes to facilitate the removal of the injection-moulded frame from the cavity. As this design consideration was not implemented, the frame became lodged within the mould. The only viable method of extraction involved applying heat to the mould to partially melt the plastic, after which the mould could be reused.

This underscores the importance of in-process quality control, as it ensures that all production steps proceed as intended. By continuously monitoring machine displays and verifying that critical parameters, such as injection pressure, are correctly configured, potential defects can be identified at an early stage before they result in material losses.

For the final quality check, it takes place by final inspection on the final product. The inspection on the product can take place in 2 ways.

- 1- Manual inspection:

This inspection happens manually through employees from the company, they check the toy cars randomly, not one by one but a random check.

2- Through an inspection machine.

The inspection machine detects if something is wrong by referencing the product to the main standards that were put for the product.

Both options have an impact on overall costs. Assuming a production volume of only 200,000 units, purchasing an inspection machine would not be cost-effective, as the fixed cost per unit would remain relatively high. Economically, such an investment would only be justified if a significantly larger quantity were produced, enabling more efficient distribution of the fixed costs. inspection. This approach, however, implies that labour costs will increase proportionally with the number of inspection hours required.

When comparing the potential cost escalation associated with both options, the analysis indicates that continuing with manual inspection performed by employees is the more financially viable solution under the current production assumptions.

However, if SpeedMass continues manufacturing the same product while maintaining the current profit margin or achieving higher margins, acquiring an inspection machine could become a viable option in the future.

Consequently, this consideration leaves SpeedMass with the alternative of manual.

An alternative solution would be to add 1.5% of the produced amount considering a failure factor. This would be due to any cause, condition, or parameter that increases the likelihood of producing defective toy cars or disrupting the process. However, a downside for this option is that the manufacturing costs increase, and so the cost of the final product increase

Conclusion

To summarise, the cost calculation for the SpeedMass toy car included material costs for parts and machine costs for processes such as injection molding, vacuum forming, and laser cutting. Assembly and packaging were also factored in, resulting in a total production cost of €3.7 per unit. While a profit margin of €1.57 per unit seems promising, it carries risks

such as underestimated costs and unaccounted expenses like shipping and returns. However, it can still be put under the average market costs of mass produced toy cars